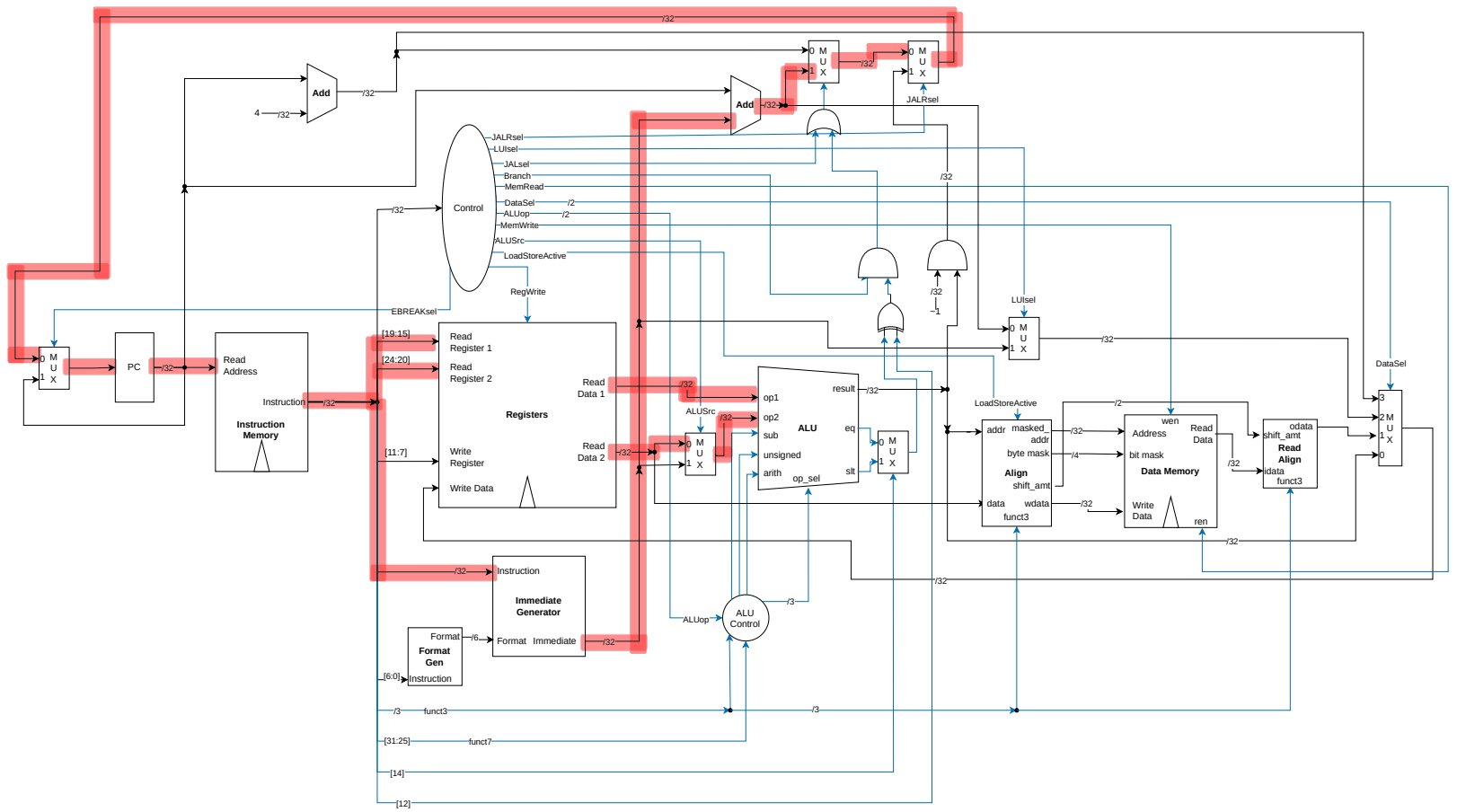
Add
4
/32
Control
JALRsel
LUIsel
JALsel
Branch
MemRead
DataSel
ALUop
MemWrite
ALUSrc
LoadStoreActive
RegWrite
EBREAKsel
PC
Read Address
Instruction
Instruction Memory
[19:15]
[24:20]
[11:7]
Read Register 1
Read Register 2
Registers
Write Register
Write Data
Read Data 1
Read Data 2
ALUSrc
op1
op2
sub
unsigned
arith
ALU
result
eq
slt
op_sel
LoadStoreActive
addr
masked addr
byte mask
Align
shift_amt
data
wdata
funct3
wen
Address
Read Data
bit mask
Data Memory
Write Data
ren
odata
shift_amt
Read Align
idata
funct3
DataSel
Instruction
Immediate Generator
Format
Immediate
Format Gen
[6:0]
ALUop
ALU Control
funct3
funct7
[31:25]
[14]
[12]